AlgoWeaver Delivery Report

Algorithm capability factory prototype - features, validation and demo

Overview

AlgoWeaver turns business materials, historical code and validation results into traceable algorithm assets. The main scenario is inventory demand forecasting, with anomaly detection and predictive maintenance examples.

Implemented capabilities

Module	Evidence
Ingestion	Markdown, text, Python, CSV, JSON and YAML with source locations and content hashes
Capability extraction	Pydantic capability cards record inputs, outputs, steps, parameters, constraints, metrics, evidence and status
Retrieval and planning	Lexical retrieval, optional Qdrant and reranking; LangGraph workflow orchestration
Generation and validation	Template, mock and replaceable LLM modes; AST, contracts, tests, MAE/RMSE and isolated execution
Capability assets	SQLite versioning, verified gate, JSON-LD export and structured repair experience
Candidate comparison	Moving-average, naive-last and weighted moving average with JSON/Markdown reports
User interfaces	Typer CLI, FastAPI and Streamlit workspace

Validation results

Cloud validation: pytest 146 passed and 1 skipped because WSL was unavailable; Ruff and Mypy passed. GitHub Actions passed lint, tests and type checks on Python 3.12 and 3.13.

Visual workspace

The workspace includes task setup, execution history, capability assets and settings. The visible flow is ingestion, extraction, retrieval, planning, validation and asset promotion. Users can enter a historical sequence and compare candidate algorithms.

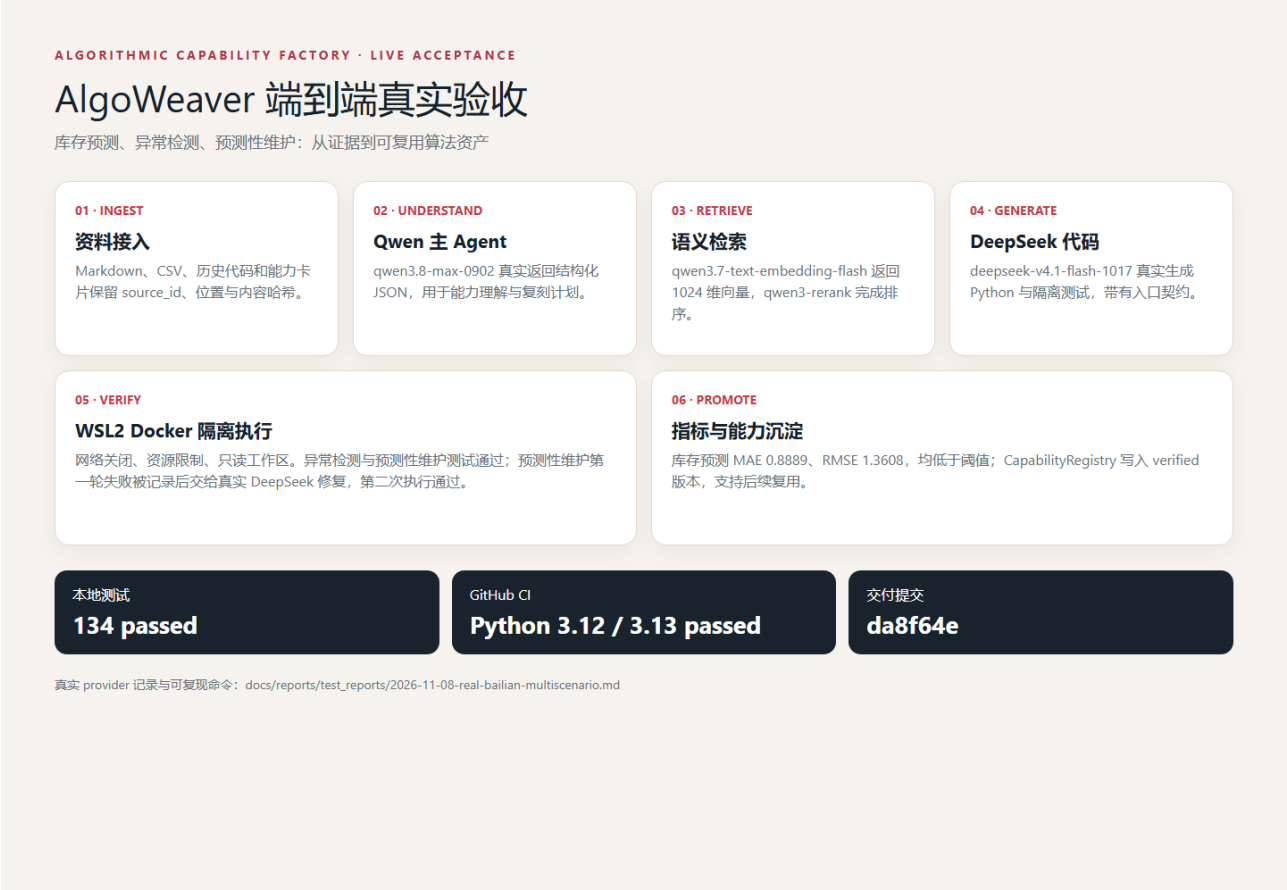


Figure 1: AlgoWeaver end-to-end acceptance flow (docs/assets/acceptance-flow.png).

How to run

uv sync --group dev --extra api --extra retrieval --extra execution --extra ui
uv run pytest -q
uv run streamlit run src/algo_weaver/ui.py --server.headless true --server.port 8512

Current boundaries

Complex PDF and Office layouts, graph visualization, plugin registration, multi-round repair, performance budgets and formal public deployment remain future extensions. External providers and Docker/WSL should be rechecked in the target environment.